

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Software Design Report

SWE 4404: SPL-II

Project Title- Dengue Alert & Ward Base management System

Team Members

Md Rufayed Ul Alam (230042121)

Sheikh Irtiaj Ahmed (230042130)

Adiba Ahsan Raiyan (230042161)

Supervised By

Ajwad Abrar Mostofa, Junior Lecturer

Department of Computer Science and Engineering

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1. Introduction

Dengue Alert & Ward-Based Management System is a mobile-first Progressive Web App (PWA) that connects citizens, ward councillors, cleanup workers and city authorities through a structured dengue hazard reporting and resolution workflow.

2. System Overview

Citizens submit dengue hazard reports with photographs, GPS coordinates and optional descriptions. The system automatically identifies the correct ward using geographic information and routes the complaint to the responsible Councillor. The Councillor verifies the complaint and assigns cleanup work to available workers. Workers submit timestamped and geo-tagged proof images after cleanup. Councillors perform final verification before closing the case.

The system also supports hotspot detection, magistrate escalation through secure links, SMS notification, and proximity-based dengue warnings.

3. Software Design Approach

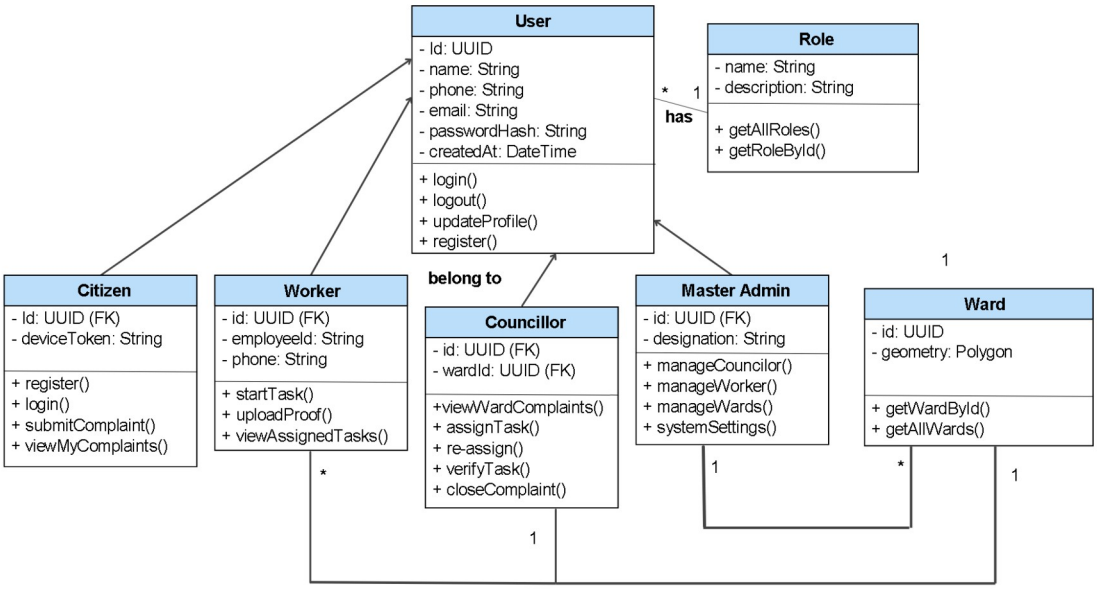
The project follows an Object-Oriented (OO) approach. Therefore, the required software design models are UML Class Diagram and Software Design Architecture Diagram.

4. UML Class Diagram

The complete UML Class Diagram has been separated into four diagrams to improve readability. Each part represents a logical group of classes.

4.1 User and Role Management

Classes: User, Citizen, Worker, Councillor, Super Admin, Role, Ward.
Purpose: Represents authentication, authorization and ward-based access control.

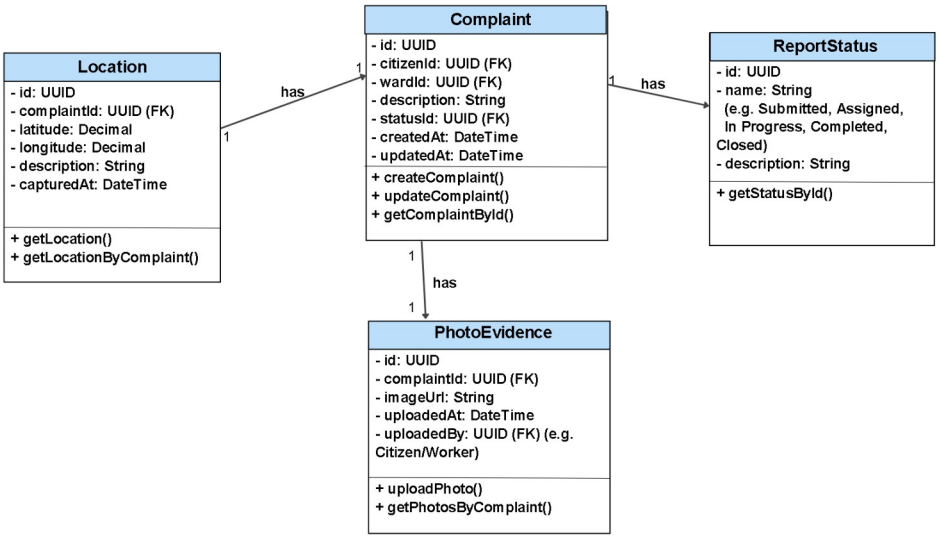


4.2 Complaint and Evidence Management

Classes: Complaint, Location, Photo Evidence, Report Status.

Purpose: Represents citizen reports, GPS information, images and complaint lifecycle.

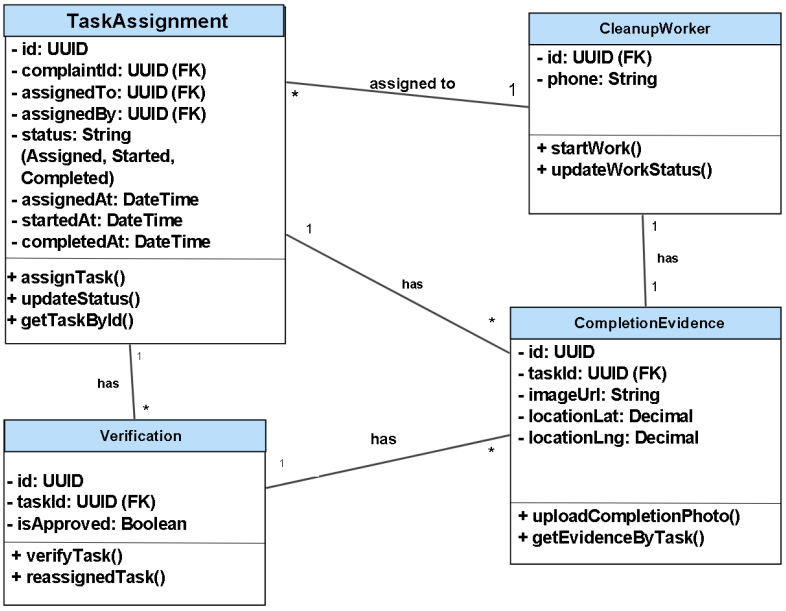
4.3



Task Assignment and Verification

Classes: Task Assignment, Cleanup Worker, Verification, Completion Evidence.

Purpose: Represents Councillor assignment, worker execution and final approval.

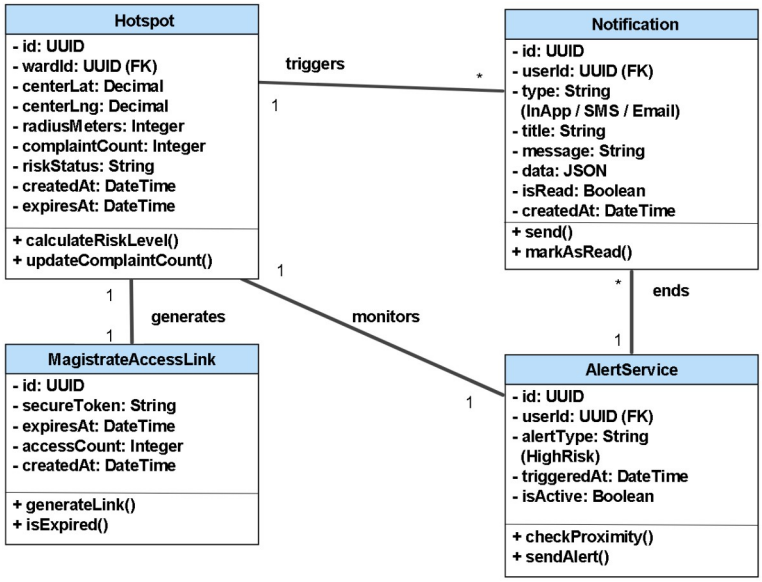


4.4 Hotspot, Notification and Escalation Management

Classes: Hotspot, Notification, Magistrate Link, Alert Service.

Purpose: Represents complaint clustering, alerts, SMS escalation and proximity warnings.

5.

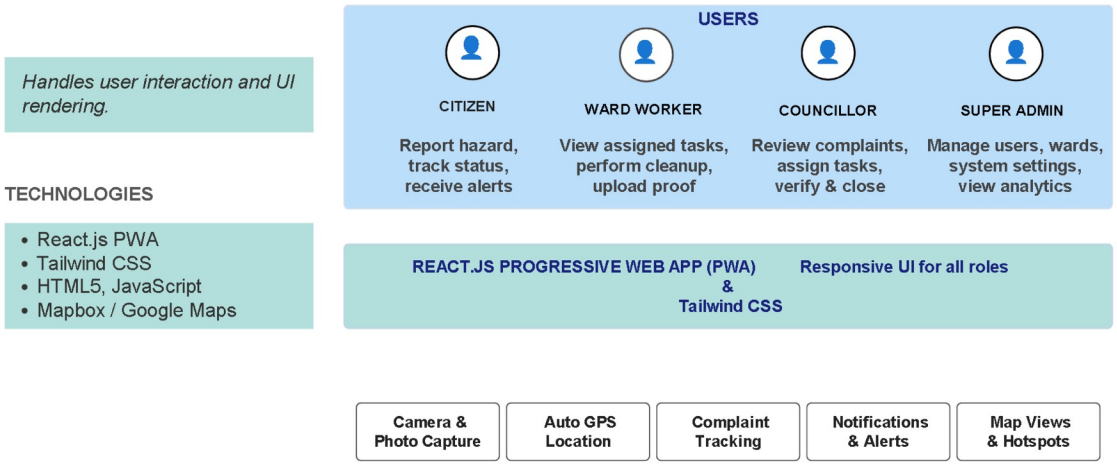


Software Design Architecture Diagram

The system follows a three-layer architecture.

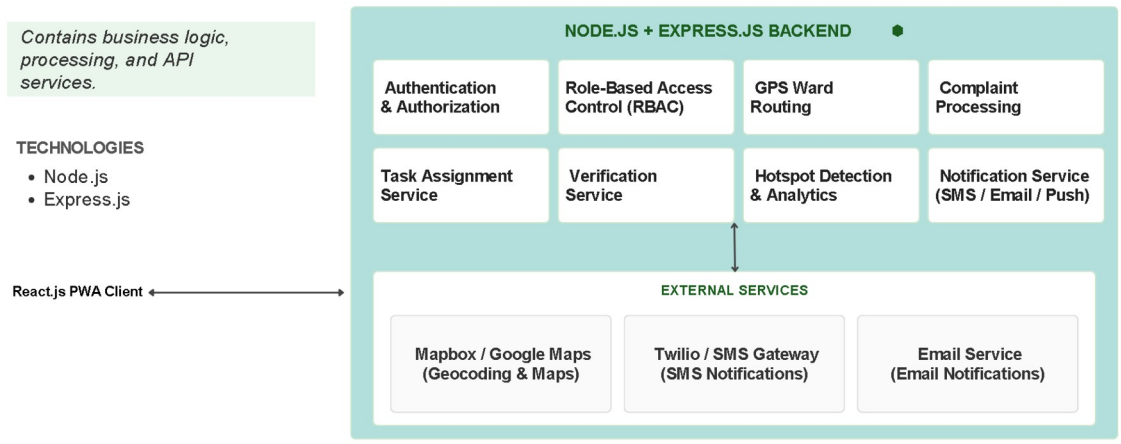
5.1 Presentation Layer

React.js PWA provides interfaces for Citizen, Worker, Councillor and Super Admin users.



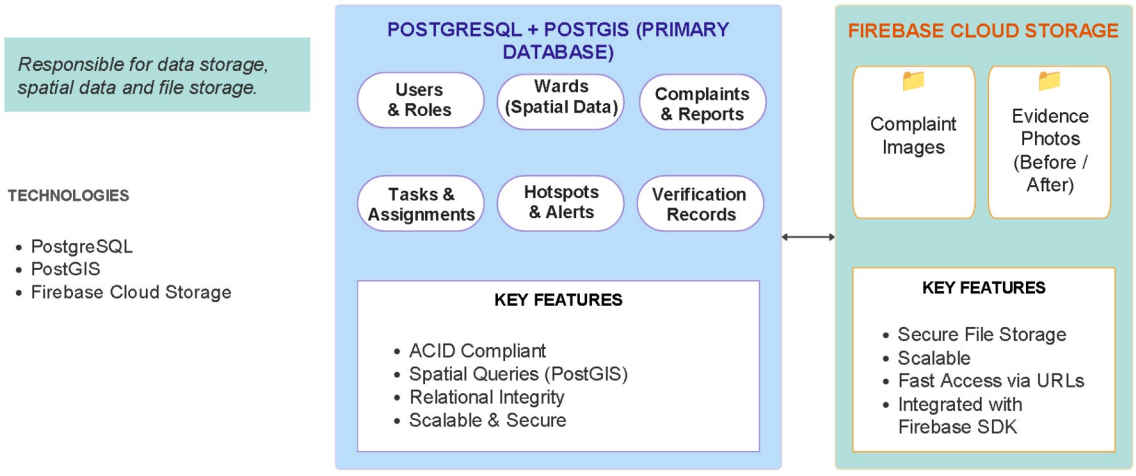
5.2 Application Layer

Node.js with Express manages authentication, routing, business rules, task assignment, notification and verification logic.



5.3 Data Layer

PostgreSQL and PostGIS manage structured and geographic data. Firebase Cloud Storage stores images.



6. Detailed Software Requirement Specification (SRS)